

Supplemental Digital Content 2

URI temporal modifier reference table — full six-state multiplier set.

This table provides the complete URI temporal modifier multipliers referenced in Section 2.5 of the main manuscript. Multipliers were derived from controlled rhinovirus-challenge data (references 9–11 in main manuscript) and ETDQ-7 meta-analyses (reference 16).

URI state	RA multiplier	PO' shift (daPa)	Equalization rate multiplier	Valsalva clearance multiplier	Per-descent relative risk
None	1.0	0	1.00	1.00	1.00
Day 1–3	2.0	+80	0.60	0.70	2.50
Day 4–7	3.5	+150	0.40	0.50	4.25
Day 8–14	1.8	+60	0.75	0.80	2.00
Day 15–21	1.3	+20	0.90	0.90	1.30
Day 22–28	1.1	+5	0.95	0.95	1.10

RA = Eustachian-tube resistance; PO' = passive Eustachian-tube opening pressure (1 daPa = 0.075 mmHg). The peak-dysfunction window (days 4–7) drives the dominant URI-state effect on per-exposure barotitis probability in the main manuscript Section 3.1.